

Math 6601 Homework 6

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Problem 1.

Set $x_i = i - 3$ for $i = 1, 2, \dots, 6$. If $q(x) = p(x) + r(x)$, then $q(x)$ will take the required values on $\{x_i\}_{i=1}^6$ provided that $r(x_6) = 31$ and $r(x_i) = 0$ if $1 \leq i < 6$. We can ensure that $r(x)$ is a polynomial passing through these points by using polynomial interpolation. In the Lagrange form, this means that

$$r(x) = \sum_{k=1}^n r(x_k) L_k(x) = r(x_6) L_6(x) = 31 \prod_{i=1}^5 \frac{x - x_i}{x_6 - x_i} = 31 \cdot \frac{x+2}{5} \cdot \frac{x+1}{4} \cdot \frac{x}{3} \cdot \frac{x-1}{2} \cdot (x-2).$$

Then

$$q(x) = p(x) + r(x) = x^4 - x^3 + x^2 - x + 1 - \frac{31}{120}(x+2)(x+1)x(x-1)(x-2)$$

is a polynomial that takes the required values.

Problem 2.

Since P_3 has degree less than or equal to 3, we can write $P_3(x) = 6x^3 + bx^2 + cx + d$ for some constants a, b, c, d . Then $P_3(0) = 0$ implies that $d = 0$, and the remaining interpolation points yield the system of equations

$$\begin{aligned} 8y &= 6 + 2b + 4c \\ 3 &= 6 + b + c \\ 2 &= 48 + 4b + 2c. \end{aligned}$$

Solving the last two equations by elimination gives $-4 = 36 + 2b \implies b = -20$, and $c = -3 - b = 17$. Thus, $8y = 6 - 40 + 68 = 34$, so $y = \frac{17}{4}$.

Problem 3.

Since $S(x)$ is a cubic spline, it must satisfy $S_0(2) = S_1(2)$, and $S'_0(2) = S'_1(2)$, and $S''_0(2) = S''_1(2)$. Then

$$a = S_1(2) = S_0(2) = 3 + 2 - 1 = 4,$$

and

$$b = S'_1(2) = S'_0(2) = 3 + 4 - 3 = 4,$$

and

$$2c = S''_1(2) = S''_0(2) = 4 - 6 = -2.$$

Thus, $a = 4$, $b = 4$, and $c = -1$.

Problem 4.

Let A be strictly diagonally dominant. Then $Ax = b$ has a unique solution.

Proof. Let R_i be the Gershgorin radius for the i th row of A , that is,

$$R_i = \sum_{j \neq i} |a_{ij}|,$$

where $A = \{a_{ij}\}$. Then, by the assumption of strict diagonal dominance, it follows that $|a_{ii}| > R_i$ for all i . Hence, for any i , we have $|0 - a_{ii}| = |a_{ii}| > R_i$, which implies that $0 \notin \{z \in \mathbb{C} : |z - a_{ii}| < R_i\} = D_i$, the Gershgorin disc for the i th row of A . Thus, 0 does not belong to any of the Gershgorin discs of A , so 0 cannot be an eigenvalue of A by the Gershgorin circle theorem. This implies that all the eigenvalues $\lambda_1, \dots, \lambda_n$ of A are nonzero. Then

$$\det(A) = \prod_{i=1}^n \lambda_i \neq 0,$$

which implies that $Ax = b$ has a unique solution. □

Problem 5.

- (a) Using the notation from the slides, we must have $a_0 = 0$, and $a_1 = 4$. The condition $S_0''(1) = 4$ forces us to take $c_0 = 2$, as $2c_0 = S_0''(1)$. From the conditions $S_1(2) = S_0(2)$, $S_1'(2) = S_0'(2)$ and $S_1''(2) = S_0''(2)$, we get

$$\begin{aligned} 4 &= a_1 = a_0 + b_0 + c_0 + d_0 = 2 + b_0 + d_0 \\ b_1 &= b_0 + 2c_0 + 3d_0 = 4 + b_0 + 3d_0 \\ 2c_1 &= 2c_0 + 6d_0 = 4 + 6d_0, \end{aligned}$$

and from the conditions $S_1(3) = 8$ and $S_1''(3) = 4$ as well as the above equations, we get

$$\begin{aligned} 8 &= a_1 + b_1 + c_1 + d_1 = 4 + b_1 + c_1 + d_1 = 10 + b_0 + 6d_0 + d_1 \\ 4 &= 2c_1 + 6d_1 = 4 + 6d_0 + 6d_1. \end{aligned}$$

These equations imply that $d_1 = -d_0$, and $b_0 = -2 - 5d_0$. Combining this with the first equation above, we have

$$4 = 2 - 2 - 5d_0 + d_0 \implies d_0 = -1.$$

Then $b_0 = 2 - d_0 = 3$, and $b_1 = 4 + b_0 + 3d_0 = 4$, and $c_1 = 2 + 3d_0 = -1$ by the first three equations. In summary, we have

$$\begin{aligned} a_0 &= 0, & b_0 &= 3, & c_0 &= 2, & d_0 &= -1, \\ a_1 &= 4, & b_1 &= 4, & c_1 &= -1, & d_1 &= 1. \end{aligned}$$

- (b) This spline is the same as the one found in Problem 3 (all the coefficients are the same, and so are the interpolation points).

Problem 6.

- (a) In this case, we again have $a_0 = 0$, and $a_1 = 4$. The condition $S_0'(1) = 3$ forces us to take $b_0 = 3$, as $S_0'(1) = b_0$. From the conditions $S_1(2) = S_0(2)$, $S_1'(2) = S_0'(2)$, and $S_1''(2) = S_0''(2)$, we get

$$\begin{aligned} 4 &= a_1 = a_0 + b_0 + c_0 + d_0 = 3 + c_0 + d_0 \\ b_1 &= b_0 + 2c_0 + 3d_0 = 3 + 2c_0 + 3d_0 \\ 2c_1 &= 2c_0 + 6d_0, \end{aligned}$$

and from the conditions $S_1(3) = 8$ and $S'_1(3) = 5$ we get

$$\begin{aligned} 8 &= a_1 + b_1 + c_1 + d_1 = 4 + b_1 + c_1 + d_1 = 7 + 3c_0 + 6d_0 + d_1 \\ 5 &= b_1 + 2c_1 + 3d_1 = 3 + 4c_0 + 9d_0 + 3d_1. \end{aligned}$$

Thus, $d_1 = 1 - 3c_0 - 6d_0$, so from the last equation we can deduce that $2 = 3 - 5c_0 - 9d_0$. From the first equation we know that $c_0 = 1 - d_0$, so we get $1 = 5 + 4d_0$, which implies that $d_0 = -1$, and $c_0 = 2$. Then $d_1 = 1$, $b_1 = 4$, and $c_1 = -1$. In summary, we have

$$\begin{aligned} a_0 &= 0, & b_0 &= 3, & c_0 &= 2, & d_0 &= -1, \\ a_1 &= 4, & b_1 &= 4, & c_1 &= -1, & d_1 &= 1. \end{aligned}$$

(b) Once again, this spline is the same as in Problems 3 and 5 (it has the same coefficients).

Problem 7.

(a) The script `p7a.m` was used to generate Table 1 and Figure 1. The output from the script is in `p7a_output.txt`.

n	e_n
4	0.438
8	1.045
16	14.387
32	5075.272

Table 1: L^∞ errors of polynomial interpolation

(b) The script `p7b.m` was used to generate Table 2 and Figure 2. The output from the script is in `p7b_output.txt`. Based on the table, it seems that $e_n = \mathcal{O}(n^{-4})$ as $n \rightarrow \infty$.

n	e_n	Order
4	2.793×10^{-1}	-
8	5.607×10^{-2}	2.32
16	3.745×10^{-3}	3.90
32	6.546×10^{-4}	2.52
64	4.013×10^{-5}	4.03
128	2.521×10^{-6}	3.99

Table 2: L^∞ errors of natural cubic spline interpolation

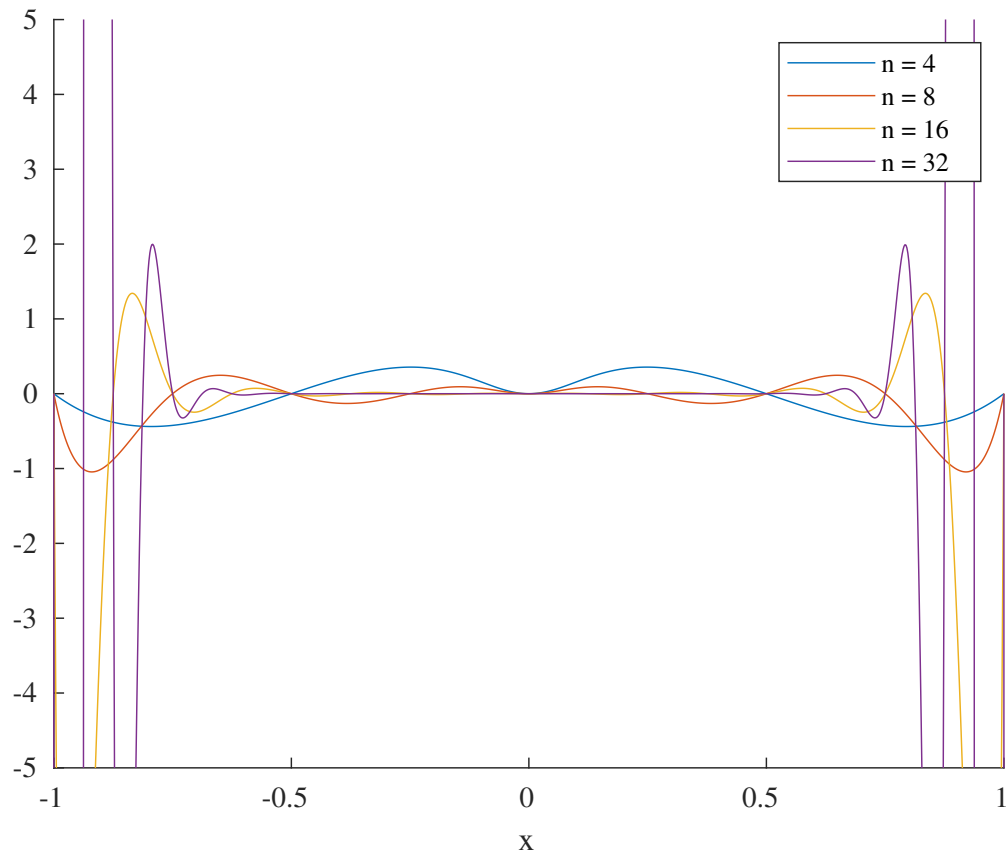


Figure 1: Plot of error function $p_n(x) - f(x)$ for polynomial interpolation

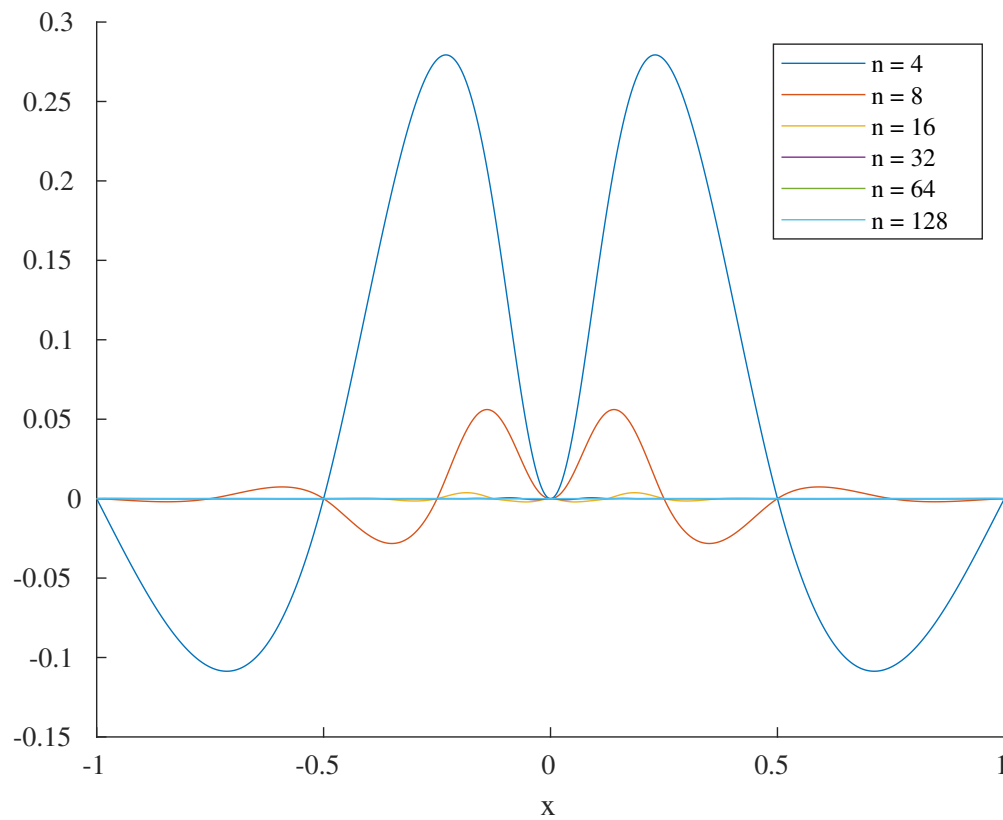


Figure 2: Plot of error function $p_n(x) - f(x)$ for natural cubic spline interpolation. Note the smaller scale compared to Figure 1.